

 rinna / **nekomata-7b**   like 7 Follow  rinna Co., Ltd. 202

 Text Generation  Transformers  PyTorch  Safetensors  5 datasets

 Japanese  English qwen custom_code  arxiv:2309.16609

 arxiv:2404.01657  License: tongyi-qianwen-license-agreement

  Train  Deploy  Use this model

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 Safetensors   Model size 7.72B params  Tensor type BF16  Files info

Inference Providers

 Text Generation

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Model tree for rinna/nekomata-7b

Base model Qwen/Qwen-7B

 Finetuned (3) [this model](#)

 Finetunes [1 model](#)

 Quantizations [3 models](#)

Datasets used to train rinna/nekomata-7b

Space using rinna/nekomata-7b 1

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qwen-nekomata  Collection

language model Collection

16 items • Updated Aug 26 • △ 3

[rinna/nekomata-7b](#)



[Overview](#)

We conduct continual pre-training of [qwen-7b](#) on **30B** tokens from a mixture of Japanese and English datasets. The continual pre-training significantly improves the model's performance on Japanese tasks. It also enjoys the following great features provided by the original Qwen model.

- The inclusive Qwen vocabulary (vocab size > 150k) enables the model to process Japanese texts much more efficiently than the previously released [youri series](#).
- The model supports a maximum sequence length of 32768.

The name [nekomata](#) comes from the Japanese word [猫又/ねこまた/Nekomata](#), which is a kind of Japanese mythical creature ([妖怪/ようかい/Youkai](#)).

- **Library**

The model was trained using code based on [EleutherAI/gpt-neox](#).

- **Model architecture**

A 32-layer, 4096-hidden-size transformer-based language model. Please refer to the [Qwen paper](#) for architecture details.

- **Continual pre-training**

The model was initialized with the [qwen-7b](#) model and continually trained on around **30B** tokens from a mixture of the following corpora

- [Japanese CC-100](#)
- [Japanese C4](#)
- [Japanese OSCAR](#)
- [The Pile](#)
- [Wikipedia](#)
- rinna curated Japanese dataset

- **Contributors**

- [Tianyu Zhao](#)
- [Akio Kaga](#)
- [Kei Sawada](#)

- **Release date**

December 21, 2023

[🔗](#) **Benchmarking**

Please refer to [rinna's LM benchmark page \(Sheet 20231221\)](#).

🔗 How to use the model

```
import torch
from transformers import AutoTokenizer, AutoModelForCausalLM

tokenizer = AutoTokenizer.from_pretrained("rinna/nekomata-7b", trust_remote_code=True)

# Use GPU with bf16
# model = AutoModelForCausalLM.from_pretrained("rinna/nekomata-7b", device_map="auto")

# Use GPU with fp16
# model = AutoModelForCausalLM.from_pretrained("rinna/nekomata-7b", device_map="auto")

# Use CPU
# model = AutoModelForCausalLM.from_pretrained("rinna/nekomata-7b", device_map="cpu")

# Automatically select device and precision
model = AutoModelForCausalLM.from_pretrained("rinna/nekomata-7b", device_map="auto")

text = "西田幾多郎は、"
token_ids = tokenizer.encode(text, add_special_tokens=False, return_tensors="pt")

with torch.no_grad():
    output_ids = model.generate(
        token_ids.to(model.device),
        max_new_tokens=200,
        min_new_tokens=200,
        do_sample=True,
        temperature=1.0,
        top_p=0.95,
        pad_token_id=tokenizer.pad_token_id,
        bos_token_id=tokenizer.bos_token_id,
        eos_token_id=tokenizer.eos_token_id
    )

output = tokenizer.decode(output_ids.tolist()[0])
```

```
print(output)
```

[Tokenization](#)

The model uses the original Qwen tokenizer. It augments the [c1100k tiktoken tokenizer](#) and has a vocabulary size of 151,936. The inclusive vocabulary helps the model to reach a better tokenization efficiency, especially for Japanese texts.

We compared the Qwen tokenizer (as used in [nekomata](#)) and the llama-2 tokenizer (as used in [youri](#)) on different text collections and found that the Qwen tokenizer achieves a much better byte2token rate (i.e. the average number of tokens produced from 1 byte of text) as following. A lower byte2token rate indicates a better tokenization efficiency.

Tokenizer	Japanese	English	Multilingual
Qwen	0.24	0.27	0.27
llama-2	0.40	0.29	0.36

[How to cite](#)

```
@misc{rinna-nekomata-7b,  
  title = {rinna/nekomata-7b},  
  author = {Zhao, Tianyu and Kaga, Akio and Sawada, Kei},  
  url = {https://huggingface.co/rinna/nekomata-7b}  
}  
  
@inproceedings{sawada2024release,  
  title = {Release of Pre-Trained Models for the {J}apanese Language},
```

```
author = {Sawada, Kei and Zhao, Tianyu and Shing, Makoto and Mitsui, Kentaro and  
booktitle = {Proceedings of the 2024 Joint International Conference on Computer  
month = {5},  
year = {2024},  
pages = {13898--13905},  
url = {https://aclanthology.org/2024.lrec-main.1213},  
note = {\url{https://arxiv.org/abs/2404.01657}}  
}
```

[References](#)

```
@software{gpt-neox-library,  
  title = {{GPT}-N: Large Scale Autoregressive Language Modeling in PyTorch},  
  author = {Andonian, Alex and Anthony, Quentin and Biderman, Stella and Black, Scott and  
  doi = {10.5281/zenodo.5879544},  
  month = {8},  
  year = {2021},  
  version = {0.0.1},  
  url = {https://www.github.com/eleutherai/gpt-neox}  
}
```

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